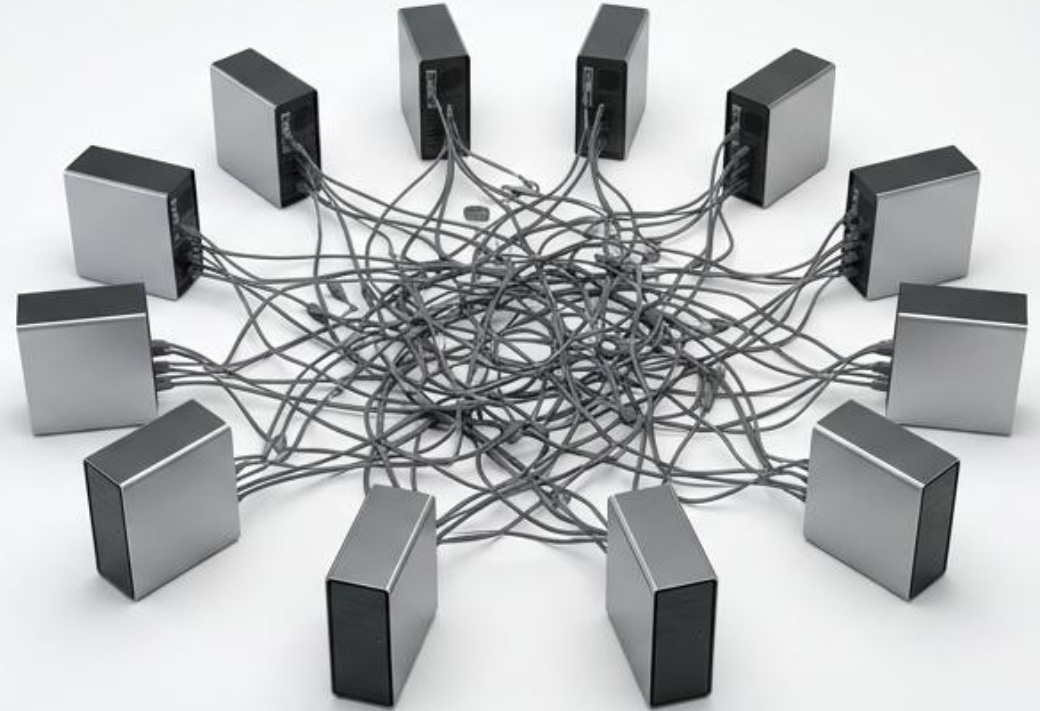




What are Network Devices?



Cables are simply the roads.





What are Network Devices?



Network devices are hardware devices used to connect computers and control data communication in a network.

They help to send, receive, filter, boost, route and translate data.



Types of Network Devices



- Modem
- Network Interface Card (NIC)
- Repeater
- Hub
- Switch
- Router
- Bridge
- Gateway



Modem



Modulation (Sending)	Demodulation (Receiving)
Converts the computer's rigid digital signals into smooth, transportable analog signals over wires.	Converts incoming analog signals back into digital data the computer's processor can read.

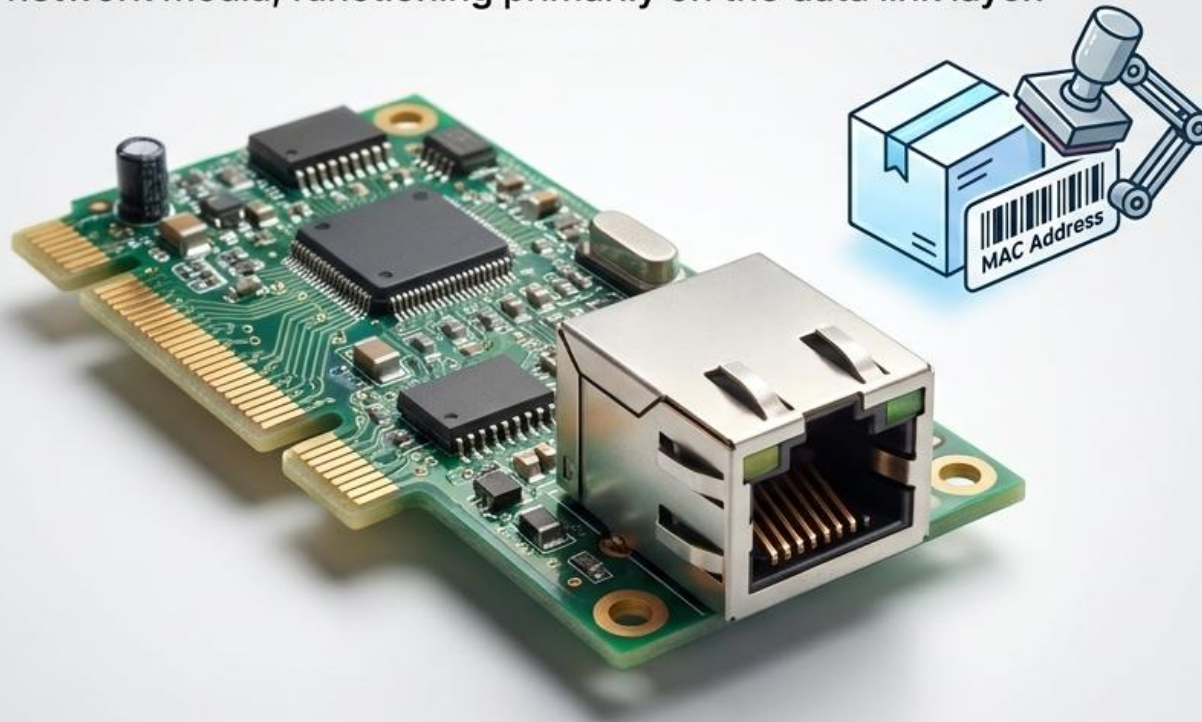


Network Interface Card (NIC)



Network Interface Card (NIC): The Passport

The physical interface that establishes a connection with network media, functioning primarily on the data link layer.



1. Identity

Stores a permanent, unique MAC (Media Access Control) address in non-volatile memory.

2. Packaging

Translates data into electrical or radio signals and attaches both source and destination MAC addresses to the outgoing packet.

3. Verification

Inspects incoming packets. If the destination MAC doesn't match the computer's identity, the packet is instantly dropped.



Repeater



Repeater: The Energy Boost

Core Problem: Attenuation. Data signals weaken, flatten, and degrade as they travel over long distances across network media.



Amplifier

Increases the amplitude (power) of the outgoing transmission back to full strength. Warning: Blindly amplifies existing noise and interference alongside the data.

Signal Regenerator

Actively processes the signal, deletes all noise and unnecessary data, and neatly retransmits a pristine, rebuilt signal, improving overall performance.



Hub



The Limitation

There is absolutely no intelligence inside a hub. Whatever signal arrives is blindly broadcast to every single device on the network **without any changes** or filtering.

The Result

Intended recipient keeps data; others discard it.

Creates massive network traffic jams.

Extends the collision domain.

Severe drop in network speeds.



Hub: How It Sends Data



Example:

Computer A wants to send data to Computer C.

Hub sends the data to B, C, D, E — everyone.
Only C accepts it; others ignore it.

Memory line: Hub shouts to everyone.



Switch



The Upgrade

A switch does not broadcast blindly. It actively reads the MAC address of the destination computer and routes the signal strictly to the specified receiver port.

The Result

Limits the collision domain.

Drastically reduces unnecessary network traffic.

Allows for full-duplex, high-speed data forwarding.

Capable of creating virtual LANs (VLANs).



Hub vs Switch



Feature	Hub	Switch
Data sending	Sends to all devices	Sends to correct device only
Intelligence	Not intelligent	Uses MAC address
Speed	Slow	Fast
Traffic	More traffic	Less traffic
Security	Less secure	More secure



Router



What is it?

A router connects two or more networks.

How it works:

- Uses IP address.
- Selects the best path for data.
- Connects home/school LAN to the internet.

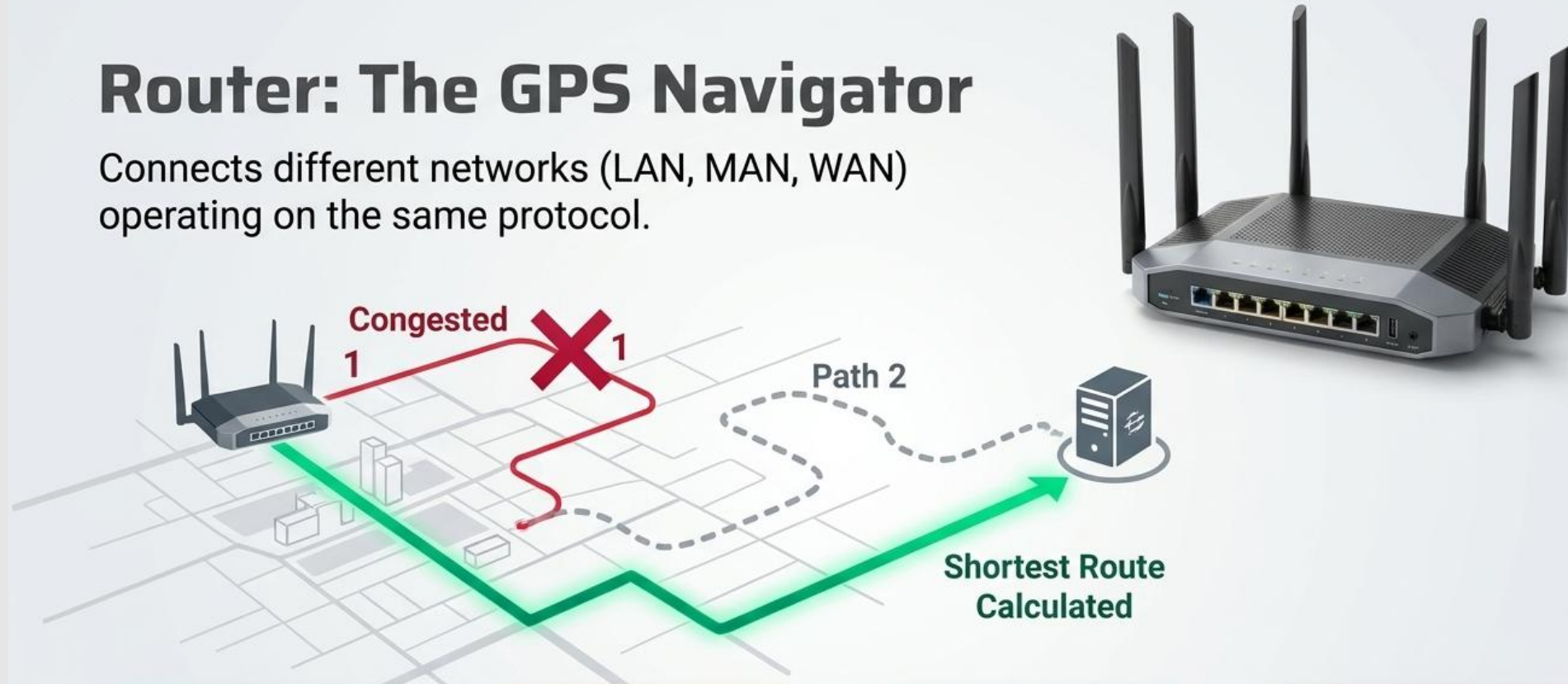


Router



Router: The GPS Navigator

Connects different networks (LAN, MAN, WAN) operating on the same protocol.



Intelligence

Calculates the fastest, shortest path for data across interconnected networks.



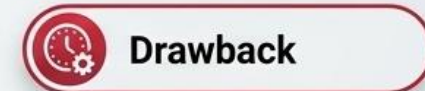
Adaptability

Uses NAT (Network Address Translation) and dynamically reroutes to bypass congestion.



Domain Control

Limits the broadcast domain and actively filters data packets.



Drawback

Operates slightly slower than a bridge due to constantly updating complex processing and routing tables.



Bridge



What is it?

A bridge connects two or more network segments of the same type.

How it works:

It checks MAC address and decides whether data should pass to another segment.

Example: connecting two computer labs.

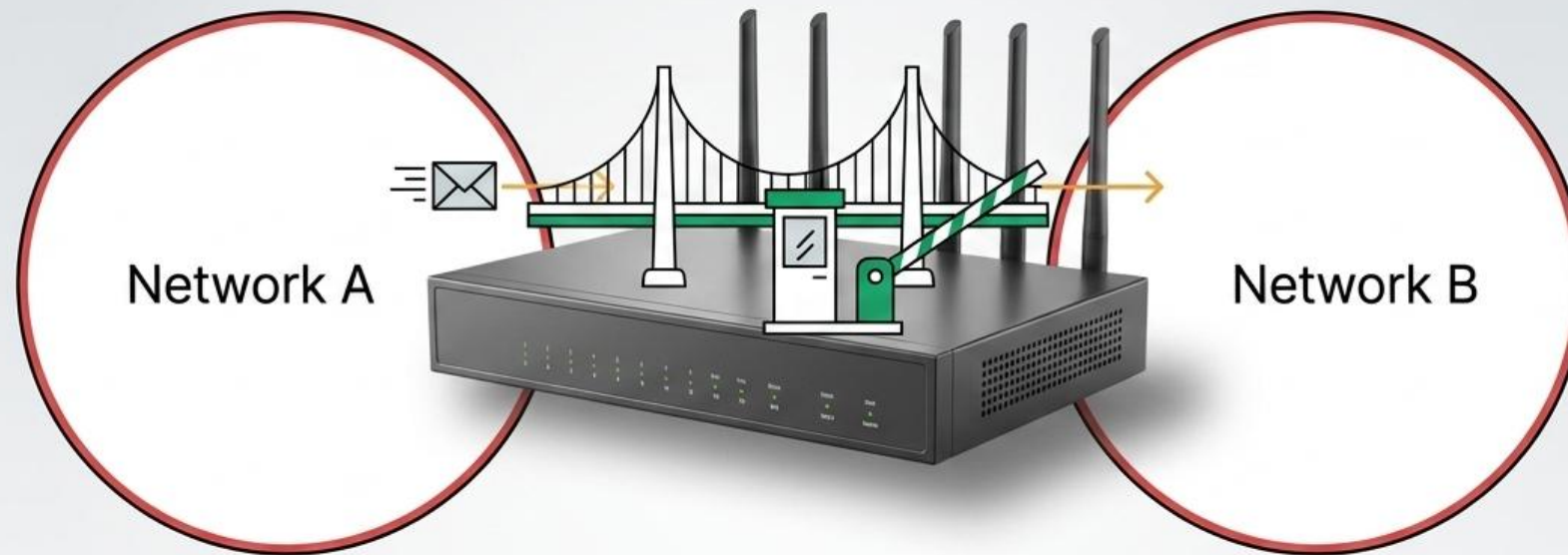


Bridge



Bridge: The Local Segmenter

Core Function: Connects multiple smaller network segments into a larger network at the Data Link layer.



How It Works:

Uses MAC addresses to control traffic movement.

It keeps local traffic local, only opening the gate for data if the destination MAC address physically resides on the other side, massively reducing overall congestion.

Bridge Typology:



1. Local Bridge:
Directly connected to the LAN.



2. Remote Bridge:
Connects globally isolated networks.



3. Wireless Bridge:
Connects multiple WLANs.



Gateway



What is it?

A gateway connects networks that use different protocols.

How it works:

It translates one protocol into another.

Memory line: Gateway = network translator.



Gateway



Gateway: The Universal Translator

Connects networks of entirely different architectures, rules, and protocols.



How It Works

- Acts as a dedicated protocol converter.
- Identifies networks using **PAT** (Protocol Address Translation).
- Sends signals strictly to target IP addresses, minimizing data collision risks.

The Trade-Offs

- **Advantage:** Unlocks ultimate interoperability between completely foreign network systems.
- **Disadvantage:** Complex installation and extra translation processing can impact overall network performance.



Final Summary



From Your Motherboard to the World

No single device builds a network. Connectivity is an orchestra of hardware—translating, amplifying, directing, and converting data millions of times a second.

